FYS4411/FYS9411: Computational Physics II

Variational Monte Carlo for Trapped Bosons

Project 1: harmonic traps, hard-sphere correlations, and one-body densities

Marius Torsheim
Department of Physics, University of Oslo

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Abstract

A variational Monte Carlo solver was implemented for trapped bosons in harmonic oscillator potentials. The code was developed first for the non-interacting oscillator, where analytical results are available in one, two, and three dimensions, and then extended to importance sampling, parameter optimization, blocking analysis, a hard-core Jastrow correlation factor, and one-body densities. Brute-force Metropolis calculations reproduced the expected variational minimum at $\alpha = 0.5$, with $E/N = 0.5$, 1.0, and 1.5 in one, two, and three dimensions. Importance sampling reproduced the same values at the exact parameter and gave finite-variance checks at $\alpha = 0.45$. A gradient optimization of α for $N = 100$ particles in three dimensions converged from $\alpha = 0.7$ to $\alpha = 0.5$ and from $E = 158.47$ to $E = 150$. The hard-core calculations with $a/a_{\mathrm{ho}} = 0.0043$ and $\beta = \gamma = 2.82843$ gave energies above the ideal reference and minima close to $\alpha = 0.5$. The radial one-body density was only weakly modified by the short-range Jastrow factor for the parameters studied. Additional feedback-driven checks include an analytic-versus-finite-difference local-energy timing comparison, committed selected CSV result files, and a pair-distance diagnostic for the hard-core correlation.

Repository: https://github.com/Torsheim/FYS4411_Project_1

I Introduction

Dilute trapped Bose gases are useful test systems for many-body methods because the confinement is simple while the correlations can still be non-trivial. In the dilute regime, the two-body physics can be modeled through a short-range repulsion or hard-sphere boundary condition. The Project 1 problem therefore starts from the exactly solvable harmonic oscillator and then adds a repulsive hard-core Jastrow factor similar to the trapped-boson calculations of DuBois and Glyde [1] and Nilsen *et al.* [2].

The goal of this report is to document both the implementation and the validation of the variational Monte Carlo (VMC) program. The non-interacting problem is used to test the local energy, Metropolis sampler, importance sampler, optimizer, and statistical analysis against exact analytical values. The interacting problem then uses the same code framework to estimate the ground-state energy and radial density of a hard-sphere Bose gas in an anisotropic trap.

The figures in this report are intentionally placed one by one in single-column format. Each plot was generated from the CSV result files by the plotting script in the repository. This makes the results easier to read in the two-column report layout and avoids oversized multi-panel figures.

II Theory

A Hamiltonian and units

I use oscillator units with $\hbar = m = \omega_\perp = 1$. For the spherical non-interacting problem in d dimensions,

$$\hat{H}_0 = \sum_{i=1}^{N} \frac{1}{2}\left(-\nabla_i^2 + r_i^2\right). \tag{1}$$

The anisotropic interacting Hamiltonian is

$$\hat{H} = \sum_{i=1}^{N} \frac{1}{2}\left(-\nabla_i^2 + x_i^2 + y_i^2 + \gamma^2 z_i^2\right) + \sum_{i<j} V_{\mathrm{int}}(r_{ij}), \tag{2}$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and

$$V_{\mathrm{int}}(r_{ij}) = \begin{cases} \infty, & r_{ij} \le a, \\ 0, & r_{ij} > a. \end{cases} \tag{3}$$

The anisotropy is $\gamma = \omega_z/\omega_\perp$. The interacting runs used $a/a_{\mathrm{ho}} = 0.0043$ and $\gamma = 2.82843$.

B Trial wave function

The trial wave function is

$$\Psi_{\mathrm{T}}(\mathbf{R};\alpha,\beta) = \prod_{i=1}^{N} g(\mathbf{r}_i;\alpha,\beta) \prod_{j<k} f(r_{jk}), \tag{4}$$

where $\mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N)$ and

$$g(\mathbf{r}_i;\alpha,\beta) = \exp\left[-\alpha(x_i^2 + y_i^2 + \beta z_i^2)\right]. \tag{5}$$

For the spherical non-interacting oscillator, $\beta = 1$. The hard-core pair factor is

$$f(r) = \begin{cases} 0, & r \le a, \\ 1 - a/r, & r > a. \end{cases} \tag{6}$$

The implementation evaluates wave-function ratios through logarithms to avoid numerical underflow for large N.

C Local energy and drift force

The local energy is

$$E_{\mathrm{L}}(\mathbf{R}) = \frac{1}{\Psi_{\mathrm{T}}(\mathbf{R})} \hat{H} \Psi_{\mathrm{T}}(\mathbf{R}). \tag{7}$$

For the non-interacting spherical oscillator, Eq. (7) gives

$$E_{\mathrm{L}}(\mathbf{R}) = \alpha d N + \left(\frac{1}{2} - 2\alpha^2\right) \sum_{i=1}^{N} r_i^2. \tag{8}$$

Averaging analytically gives

$$\frac{\langle E \rangle}{N} = d\left(\frac{\alpha}{2} + \frac{1}{8\alpha}\right), \tag{9}$$

with minimum at $\alpha = 1/2$ and exact energy $E/N = d/2$.

The drift force for importance sampling is

$$\mathbf{F}_i = 2\frac{\nabla_i \Psi_{\mathrm{T}}}{\Psi_{\mathrm{T}}} = 2\nabla_i \ln \Psi_{\mathrm{T}}. \tag{10}$$

For the non-interacting spherical Gaussian this becomes $\mathbf{F}_i = -4\alpha \mathbf{r}_i$.

For the interacting calculation I write $f(r) = \exp[u(r)]$, where

$$u(r) = \ln(1 - a/r), \qquad r > a. \tag{11}$$

The derivatives are

$$u'(r) = \frac{a}{r(r-a)}, \tag{12}$$

$$u''(r) = -\frac{a(2r-a)}{r^2(r-a)^2}. \tag{13}$$

For particle k,

$$\nabla_k \ln \Psi_{\mathrm{T}} = \nabla_k \ln \phi(\mathbf{r}_k) + \mathbf{C}_k, \qquad \mathbf{C}_k = \sum_{j \ne k} \frac{\mathbf{r}_k - \mathbf{r}_j}{r_{kj}} u'(r_{kj}). \tag{14}$$

The final expression used in the code is

$$\frac{\nabla_k^2 \Psi_{\mathrm{T}}}{\Psi_{\mathrm{T}}} = \frac{\nabla_k^2 \phi(\mathbf{r}_k)}{\phi(\mathbf{r}_k)} + 2\frac{\nabla_k \phi(\mathbf{r}_k)}{\phi(\mathbf{r}_k)} \cdot \mathbf{C}_k + \mathbf{C}_k \cdot \mathbf{C}_k \tag{15}$$

$$+ \sum_{j \ne k} \left[u''(r_{kj}) + \frac{2}{r_{kj}} u'(r_{kj})\right], \tag{16}$$

where the last term is written for three dimensions. The interacting local energy is therefore

$$E_{\mathrm{L}}(\mathbf{R}) = -\frac{1}{2} \sum_{k=1}^{N} \frac{\nabla_k^2 \Psi_{\mathrm{T}}}{\Psi_{\mathrm{T}}} + \frac{1}{2} \sum_{k=1}^{N} (x_k^2 + y_k^2 + \gamma^2 z_k^2), \tag{17}$$

for configurations satisfying all $r_{ij} > a$. Configurations with $r_{ij} \le a$ are assigned zero wave-function weight and rejected.

III Numerical method

The brute-force sampler uses particle-by-particle Metropolis updates with symmetric uniform displacements. Therefore, the acceptance probability is simply

$$A(\mathbf{R} \to \mathbf{R}') = \min\left(1, \frac{|\Psi_{\mathrm{T}}(\mathbf{R}')|^2}{|\Psi_{\mathrm{T}}(\mathbf{R})|^2}\right), \tag{18}$$

following the Metropolis algorithm [3]. Importance sampling uses the Langevin proposal

$$\mathbf{r}_i' = \mathbf{r}_i + D\mathbf{F}_i \Delta t + \sqrt{\Delta t}\, \boldsymbol{\eta}, \tag{19}$$

with diffusion constant $D = 1/2$ and normally distributed noise. The acceptance ratio then includes the Green's-function correction of the Metropolis–Hastings algorithm [4].

The VMC estimate of the energy is the sample mean

$$\langle E \rangle \approx \frac{1}{M} \sum_{m=1}^{M} E_{\mathrm{L}}(\mathbf{R}_m). \tag{20}$$

The gradient used for optimizing α is estimated as

$$\frac{\partial \langle E \rangle}{\partial \alpha} = 2\left(\langle E_{\mathrm{L}} O_\alpha \rangle - \langle E_{\mathrm{L}} \rangle \langle O_\alpha \rangle\right), \tag{21}$$

where $O_\alpha = \partial \ln \Psi_{\mathrm{T}} / \partial \alpha$. Correlated-sample uncertainty was studied with blocking, following Flyvbjerg and Petersen [5].

The code was organized as a small Python package with modules for the wave function, local energy, samplers, optimization, statistics, and density estimation. Unit tests covered the most important formulae and sampling utilities, and the test suite passed before the production runs.

IV Results and discussion

A Brute-force non-interacting benchmarks

Figures 1, 2, and 3 show the brute-force Metropolis energy per particle as a function of α in one, two, and three dimensions. All three plots have a minimum at $\alpha = 0.5$, in agreement with Eq. (9). Figures 4, 5, and 6 show the corresponding acceptance rates. The acceptance decreases as α increases because the Gaussian trial function becomes narrower, and it decreases with dimension because the same proposal scale changes more coordinates.

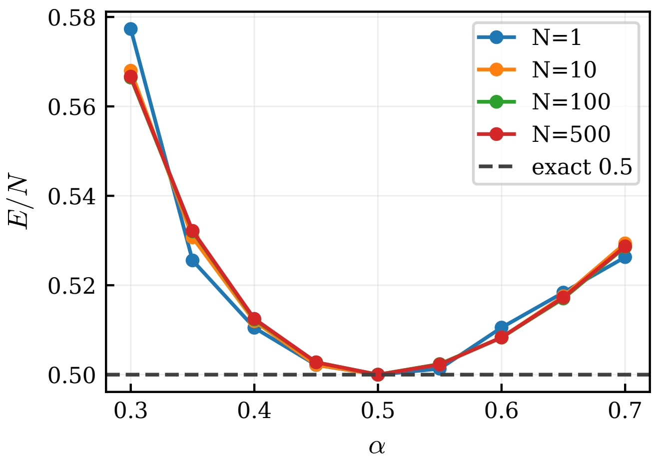

Figure 1: Brute-force Metropolis energy per particle in one dimension. The dashed line is the exact value $E/N = 0.5$.

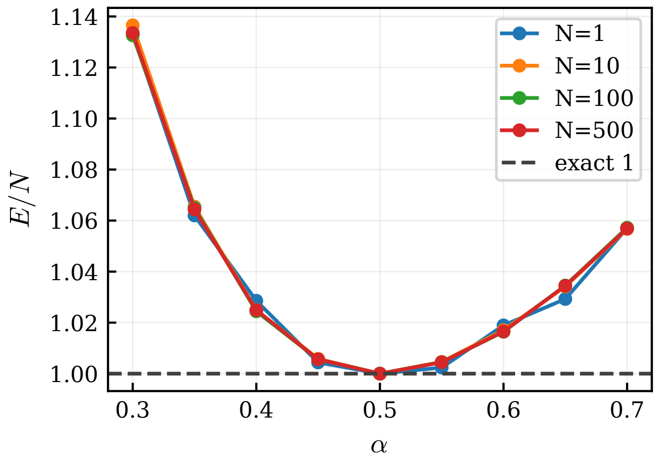

Figure 2: Brute-force Metropolis energy per particle in two dimensions. The dashed line is the exact value $E/N = 1.0$.

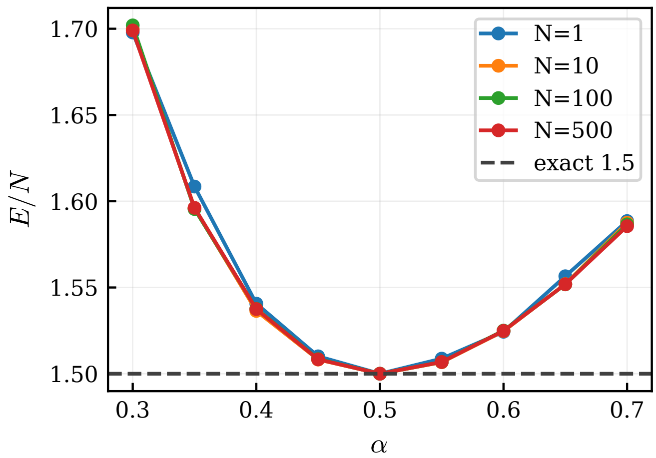

Figure 3: Brute-force Metropolis energy per particle in three dimensions. The dashed line is the exact value $E/N = 1.5$.

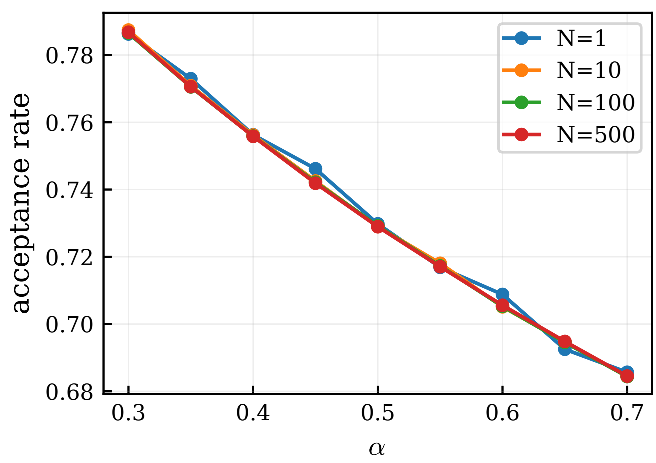

Figure 4: Acceptance rate for the one-dimensional brute-force runs.

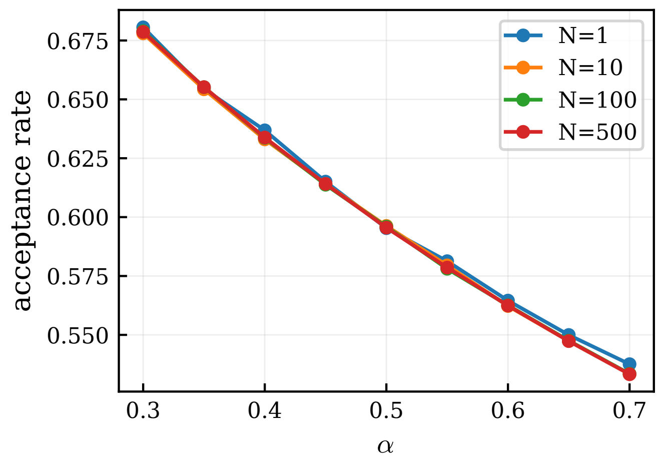

Figure 5: Acceptance rate for the two-dimensional brute-force runs.

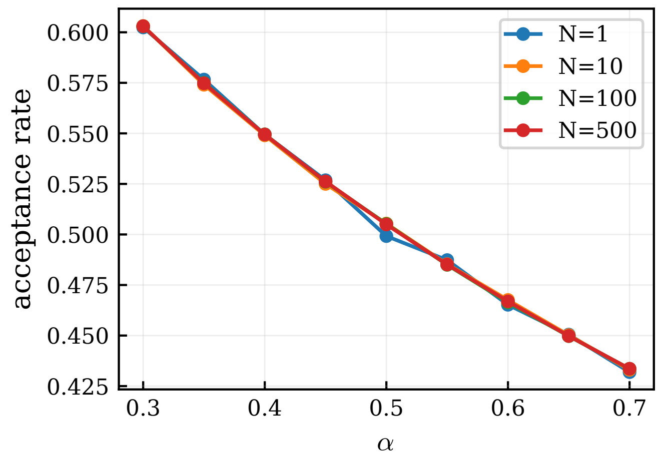

Figure 6: Acceptance rate for the three-dimensional brute-force runs.

Table 1 lists the validation points at $\alpha = 0.5$. The energy is exactly constant in these runs because Eq. (8) loses its coordinate-dependent term at $\alpha = 1/2$.

Table 1: Non-interacting brute-force validation at $\alpha = 0.5$. The exact energy per particle is $d/2$ in d dimensions.

d	N	E/N	exact	acc.
1	1	0.500000	0.5	0.730
1	10	0.500000	0.5	0.729
1	100	0.500000	0.5	0.729
1	500	0.500000	0.5	0.729
2	1	1.000000	1.0	0.595
2	10	1.000000	1.0	0.596
2	100	1.000000	1.0	0.596
2	500	1.000000	1.0	0.596
3	1	1.500000	1.5	0.499
3	10	1.500000	1.5	0.505
3	100	1.500000	1.5	0.505
3	500	1.500000	1.5	0.505

The project also asked for an analytic-versus-numerical local-energy comparison. Table 2 compares the analytic expression with a finite-difference kinetic-energy calculation for the same sampled system. The energies agree to numerical precision, while the finite-difference estimator

Table 2: Analytic and finite-difference local-energy comparison for the non-interacting harmonic oscillator. The run used $N = 10$, $d = 3$, $\alpha = 0.45$, and 3000 production cycles. The two estimators agree within numerical precision, while the analytic expression is substantially faster.

Method	E	stderr	time [s]	rel. spee
Analytic	15.06976703	0.00687	0.676	1.00
Numerical	15.06976702	0.00687	2.854	4.22× slow

Absolute energy difference: 1.18e-08.

is slower because it evaluates shifted wave functions for every coordinate of every particle.

B Importance sampling

Importance sampling was first tested at $\alpha = 0.5$. Figures 7, 8, and 9 show that the energy is exactly the known value in all dimensions. The corresponding acceptance rates in Figures 10, 11, and 12 decrease smoothly as the time step increases. These figures are best interpreted as exact-wave-function validation tests.

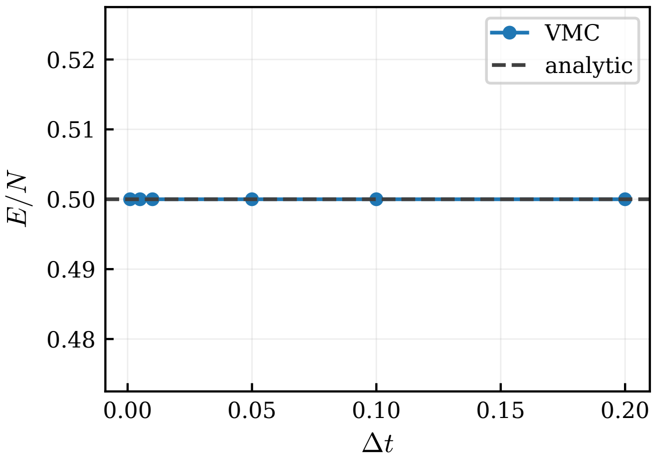

Figure 7: Importance-sampling energy in one dimension at $\alpha = 0.5$.

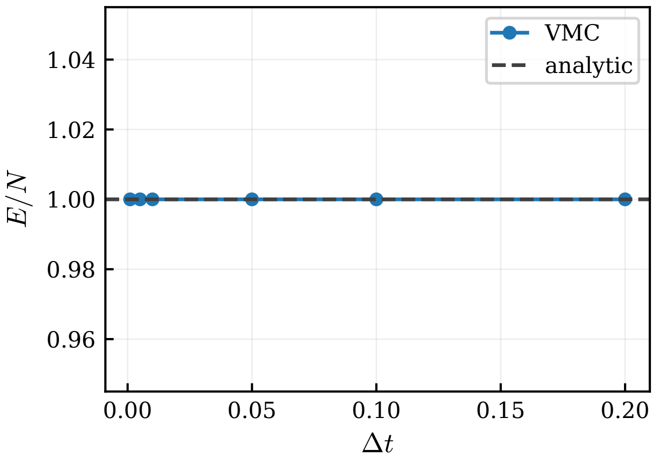

Figure 8: Importance-sampling energy in two dimensions at $\alpha = 0.5$.

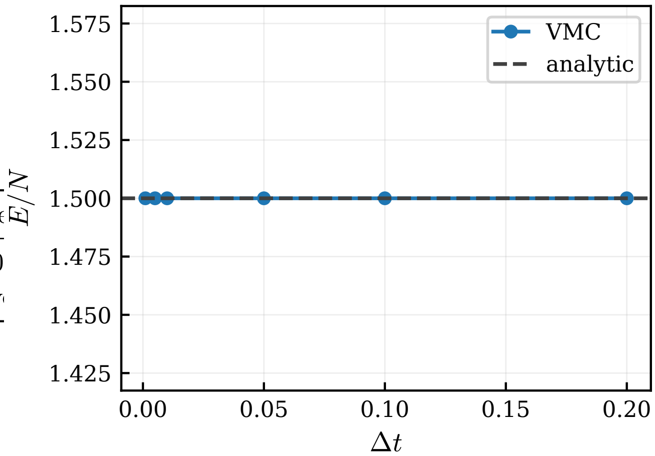

Figure 9: Importance-sampling energy in three dimensions at $\alpha = 0.5$.

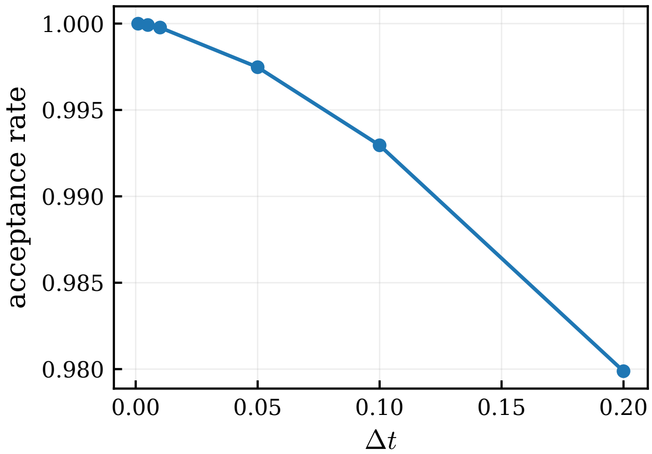

Figure 10: Importance-sampling acceptance rate in one dimension at $\alpha = 0.5$.

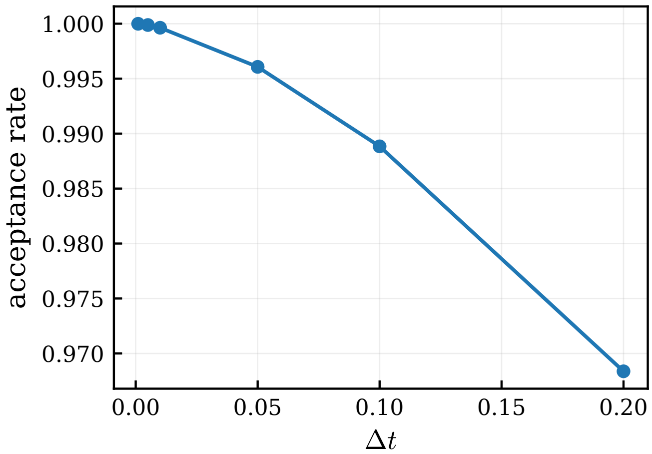

Figure 11: Importance-sampling acceptance rate in two dimensions at $\alpha = 0.5$.

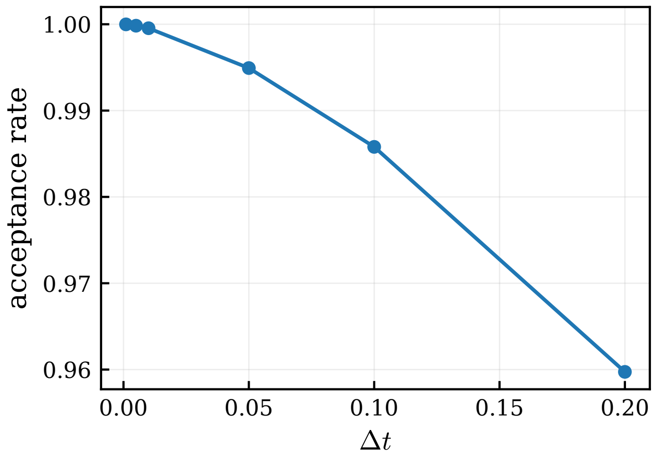

Figure 12: Importance-sampling acceptance rate in three dimensions at $\alpha = 0.5$.

The same time-step study was repeated at $\alpha = 0.45$, where the local energy has nonzero variance. Figures 13, 14, and 15 show that the energies fluctuate around the analytical variational value from Eq. (9). Figures 16, 17, and 18 show the same monotonic acceptance-rate trend. Table 3 gives selected three-dimensional numbers from the $\alpha = 0.45$ run.

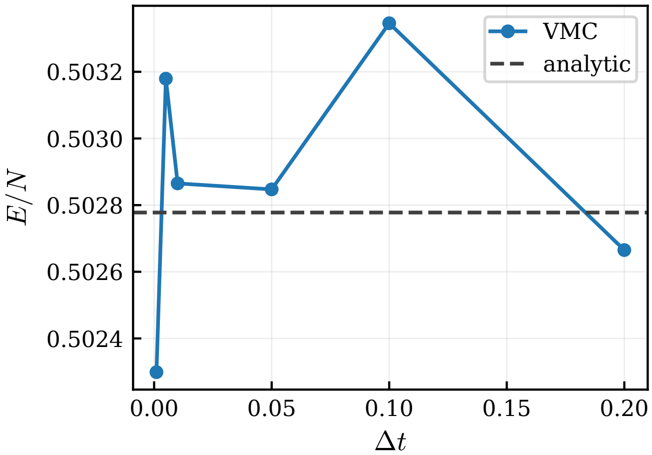

Figure 13: Importance-sampling energy in one dimension at $\alpha = 0.45$. The dashed line is the analytical variational value.

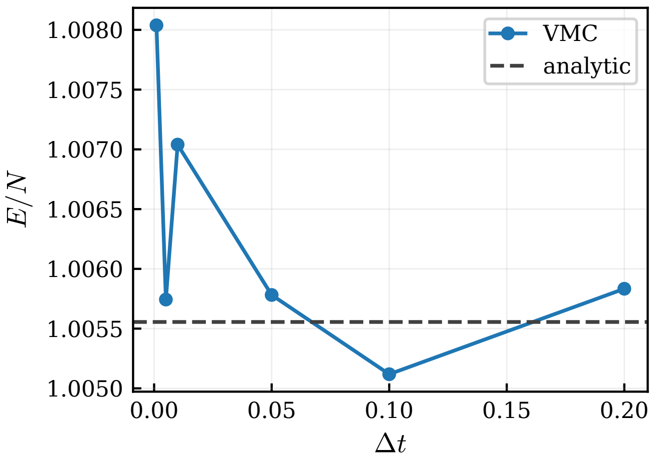

Figure 14: Importance-sampling energy in two dimensions at $\alpha = 0.45$. The dashed line is the analytical variational value.

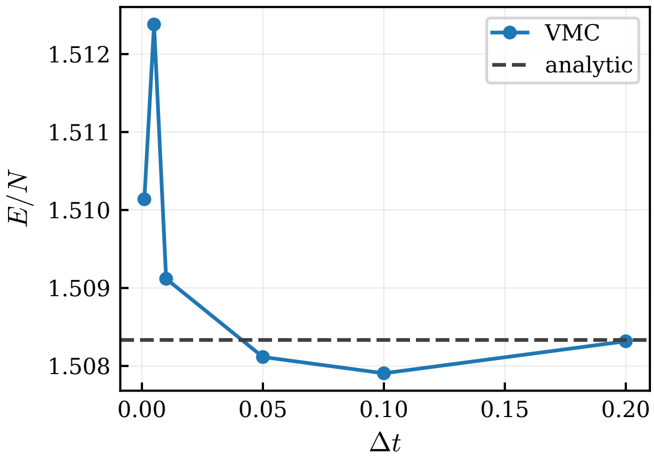

Figure 15: Importance-sampling energy in three dimensions at $\alpha = 0.45$. The dashed line is the analytical variational value.

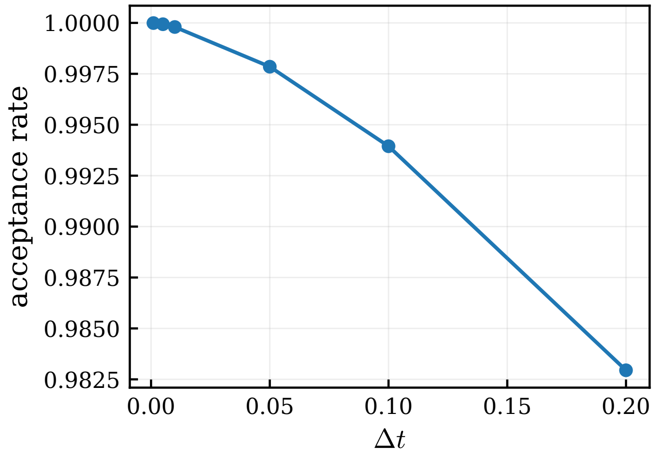

Figure 16: Importance-sampling acceptance rate in one dimension at $\alpha = 0.45$.

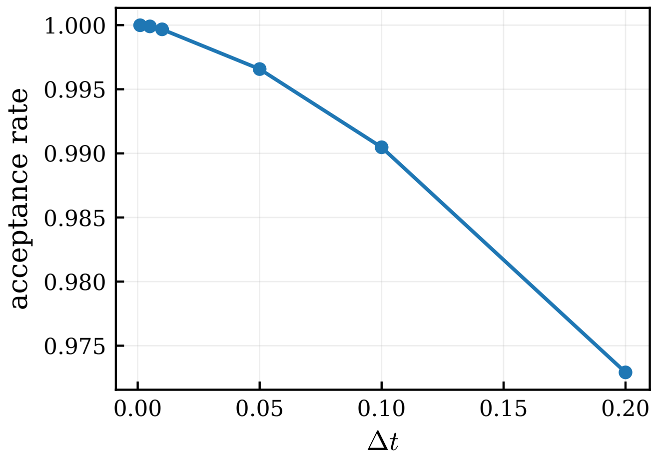

Figure 17: Importance-sampling acceptance rate in two dimensions at $\alpha = 0.45$.

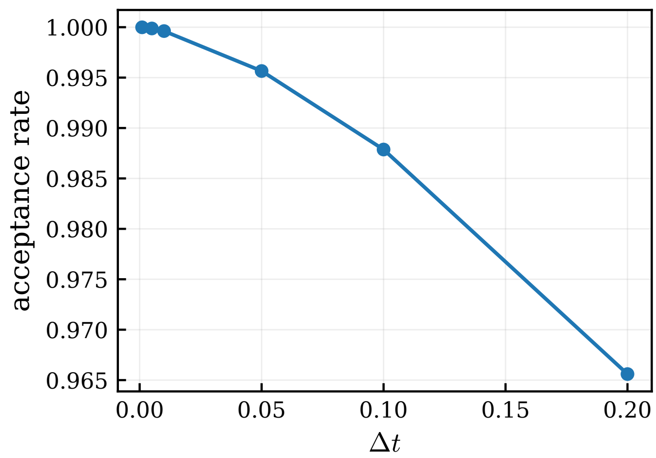

Figure 18: Importance-sampling acceptance rate in three dimensions at $\alpha = 0.45$.

Table 3: Importance-sampling time-step check for $N = 100$, $d = 3$, and $\alpha = 0.45$. The last column is the deviation from the analytic variational value.

Δt	E/N	acc.	$\Delta(E/N)$
0.001	1.510138	1.0000	+1.80e-03
0.010	1.509118	0.9996	+7.85e-04
0.050	1.508115	0.9957	-2.18e-04
0.100	1.507906	0.9879	-4.27e-04
0.200	1.508317	0.9656	-1.68e-05

C Optimization of the variational parameter

The optimization run for $N = 100$ and $d = 3$ is summarized in Table 4. Figure 19 shows that α moves smoothly from 0.7 toward the analytical optimum 0.5. Figure 20 shows the corresponding decrease in energy toward the exact value $E = 150$, and Figure 21 shows that the gradient magnitude decreases over several orders of magnitude. The optimized run therefore confirms both the gradient estimator in Eq. (21) and the chosen learning rate.

Table 4: Initial and final values of the α optimization for $N = 100$ particles in three dimensions.

step	α	E	$\lvert\partial E/\partial\alpha\rvert$
initial	0.700000	158.470392	71.6
final	0.500000	150.000000	2.4e-05

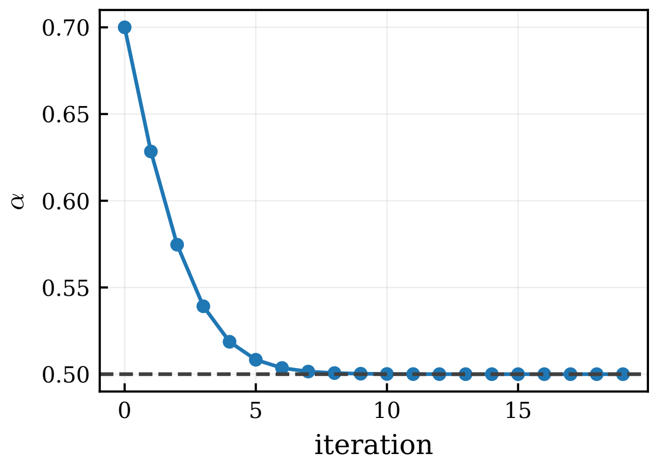

Figure 19: Optimization path for the variational parameter α for $N = 100$ particles in three dimensions.

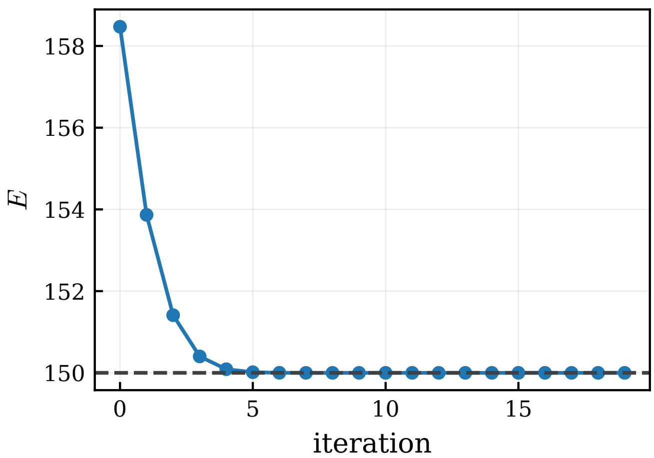

Figure 20: Energy during the α optimization. The dashed line is the exact non-interacting energy $E = 150$.

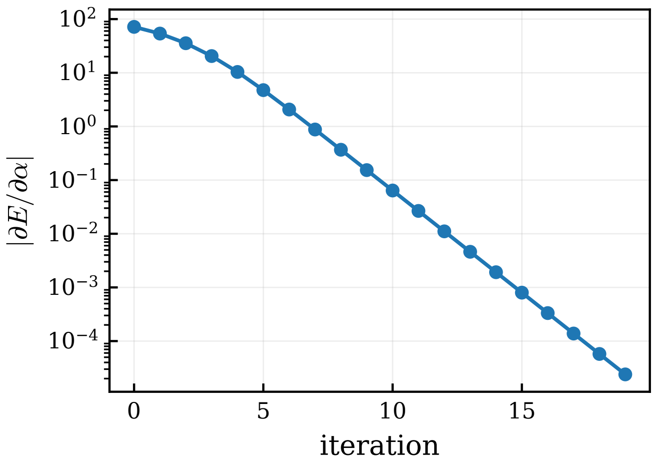

Figure 21: Magnitude of the energy gradient during the α optimization.

D Blocking and statistical errors

Blocking was applied to a finite-variance three-dimensional run with $N = 100$ and $\alpha = 0.45$. Figure 22 shows that the estimated standard error increases with block size before becoming noisy at the largest block sizes. This is the expected behavior for correlated Monte Carlo samples: small blocks underestimate the uncertainty, while very large blocks leave too few independent block averages. Table 5 lists representative blocking levels and the bootstrap and final blocking estimates.

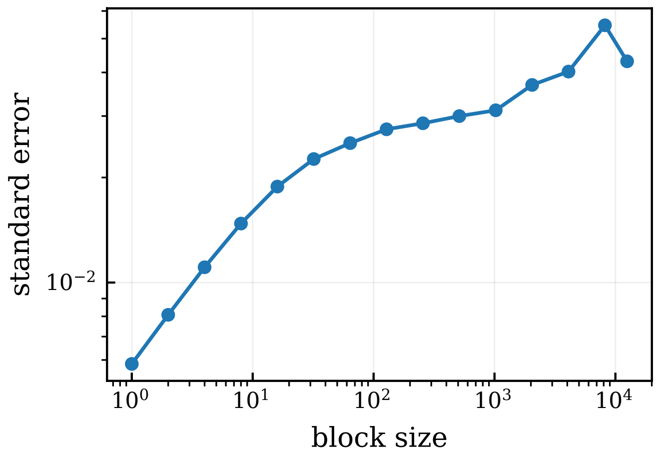

Figure 22: Blocking analysis for the $N = 100$, $d = 3$, $\alpha = 0.45$ local-energy samples.

Table 5: Selected blocking estimates for the $N = 100$, $d = 3$, $\alpha = 0.45$ run, together with the bootstrap and final blocking errors from the summary file.

block size	blocks	error
1	50000	0.00585
16	3125	0.01884
128	390	0.02749
1024	48	0.03116
8192	6	0.05460
bootstrap	–	0.00585
final block	–	0.04305

E Hard-core interaction

The interacting hard-core runs used $a/a_{\mathrm{ho}} = 0.0043$ and $\beta = \gamma = 2.82843$. Figures 23, 24, and 25 show the interacting energy per particle for $N = 10$, 50, and 100. In all cases the interacting energy lies above the ideal anisotropic reference and has a minimum near $\alpha = 0.5$. Table 6 collects the minimum values on the scanned grid. The excess energy per particle grows with N, which is consistent with stronger crowding in the trap and a larger effect of the hard-core correlations.

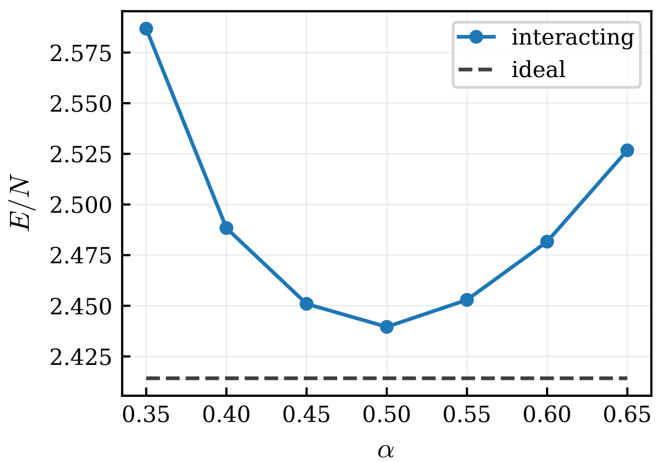

Figure 23: Interacting hard-core energy per particle for $N = 10$ particles.

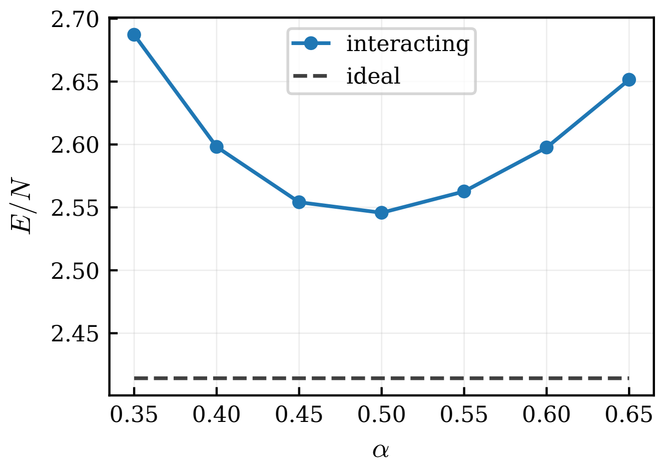

Figure 24: Interacting hard-core energy per particle for $N = 50$ particles.

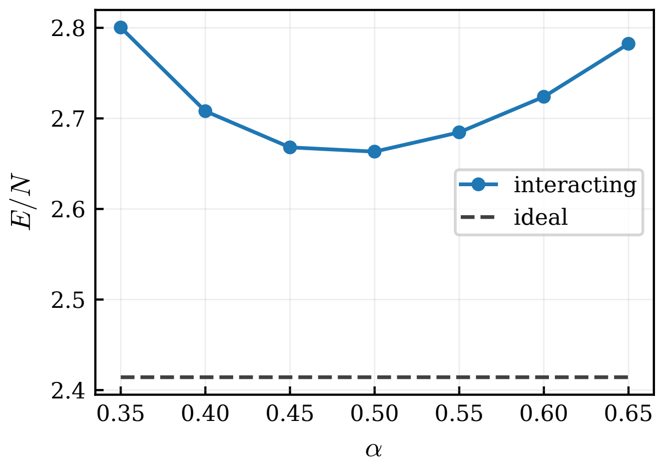

Figure 25: Interacting hard-core energy per particle for $N = 100$ particles.

Table 6: Interacting hard-core minima from the scanned α grid. Energies are per particle and the ideal reference uses the anisotropic non-interacting ground state.

N	α	E/N	ideal	excess	acc.
10	0.50	2.439590	2.414215	0.025375	0.996
50	0.50	2.545742	2.414215	0.131527	0.993
100	0.50	2.663318	2.414215	0.249103	0.991

The reference calculations of DuBois and Glyde considered the same Rb hard-sphere diameter in anisotropic traps and compared VMC energies with Gross–Pitaevskii values over a much larger particle-number range [1]. Since the present report uses only the assignment grid $N = 10, 50, 100$ and keeps only α variational, a point-by-point literature table is not directly available from those plots. Table 7 therefore reports the small-N interaction strength Na/a_{ho} and the excess over the ideal anisotropic trap as the quantitative benchmark context.

F One-body densities

The one-body density was measured with and without the Jastrow factor. Figure 26 shows the radial density $\rho(r)$,

Table 7: Interacting-run benchmark context. The published DuBois–Glyde benchmark for the same Rb hard-sphere diameter focuses on anisotropic-trap VMC values and GP comparisons over a much larger particle-number range, so the present $N = 10, 50, 100$ data are used mainly as a small-N code benchmark. The excess column is measured relative to the ideal anisotropic value $E_0/N = (2 + \gamma)/2 = 2.414215$.

N	Na/a_{ho}	α_{min}	E/N	excess
10	0.043	0.50	2.439590	0.025375
50	0.215	0.50	2.545742	0.131527
100	0.430	0.50	2.663318	0.249103

while Figure 27 shows the radial probability distribution $4\pi r^2\rho(r)$. The probability distribution is easier to interpret visually because the shell-volume factor suppresses the origin and reveals the most probable radius. Table 8 lists the peak positions. The two curves are close, indicating that the short-range Jastrow factor mainly affects pair correlations and only weakly changes the one-body density for the hard-core radius used here.

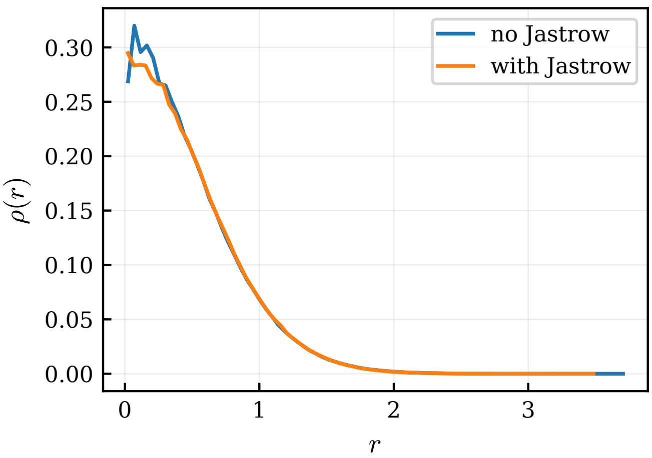

Figure 26: Radial one-body density with and without the Jastrow factor.

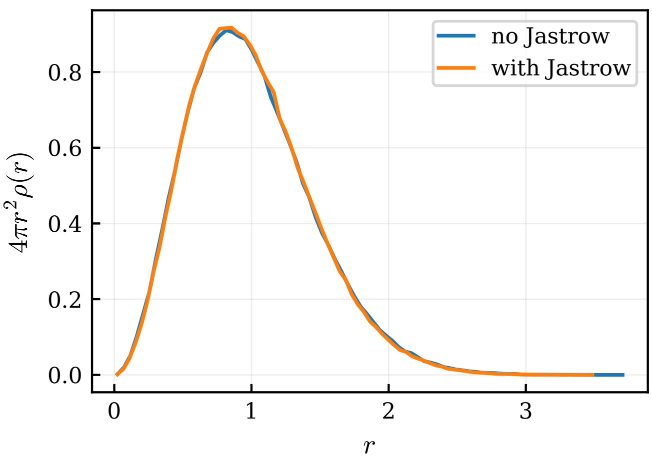

Figure 27: Radial probability distribution $4\pi r^2\rho(r)$ with and without the Jastrow factor.

Table 8: Peak positions of the radial probability distributions with and without the Jastrow factor.

case	r_{peak}	max $4\pi r^2\rho(r)$
no Jastrow	0.816	0.910
with Jastrow	0.855	0.917

Table 9: Pair-distance diagnostics for the density run. The hard-core radius is small, $a/a_{\text{ho}} = 0.0043$, so the one-body density changes weakly; the pair-distance distribution is a more direct check that the Jastrow factor affects short-range configurations.

Case	mean r_{ij}	median r_{ij}	min. sampled	acc.
No Jastrow	1.4014	1.3297	0.0246	0.671
With Jastrow	1.4134	1.3426	0.0109	0.677

To make the short-range role of the Jastrow factor more visible, Figure 28 shows the corresponding pair-distance distribution and Table 9 summarizes the mean, median, and minimum sampled pair distances. The effect is still modest because $a/a_{\text{ho}} = 0.0043$ is very small, but the diagnostic is more directly connected to pair correlations than the one-body density alone.

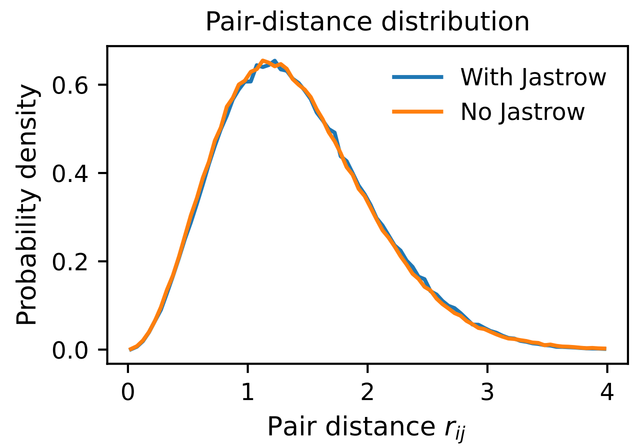

Figure 28: Pair-distance probability density with and without the hard-core Jastrow factor.

V Conclusion

The VMC implementation reproduces the exact non-interacting harmonic oscillator results in one, two, and three dimensions. The brute-force sampler gives the expected variational minimum at $\alpha = 0.5$, while the importance sampler gives the same exact results at $\alpha = 0.5$ and sensible finite-variance estimates at $\alpha = 0.45$. The optimization routine converges cleanly to the analytical optimum when the learning rate is chosen small enough. Blocking analysis shows that naive error estimates can underestimate the uncertainty when samples are correlated.

For the hard-core Bose gas, the interacting energies are consistently above the ideal reference and increase with particle number. The scanned minima remain near $\alpha = 0.5$, showing that the harmonic trap still dominates the one-body scale for the chosen parameters. The radial densities with and without the Jastrow factor are very similar, which is reasonable because the hard-core radius is small compared with the oscillator length. The feedback-driven additions close the main reproducibility gaps by committing the selected CSV result files, adding the analytic-versus-numerical timing table, and adding a pair-distance diagnostic. Future improvements would include longer interacting runs, a finer α grid, optimization of both α and β, and a fully normalized inhomogeneous pair-correlation function $g(r)$ for the hard-core system.

A Reproducibility and plotting

The report figures and tables were generated from the CSV files in `data/results/` using

```
python scripts/plot_report_single_column.py
```

The script writes vector PDF figures to `report/figures/` and table snippets to `report/tables/`. The matplotlib base font size is 10 pt, matching the body text of the report, and each figure is sized for inclusion at one-column width. The main result files used in the report are stored in `data/results/`; they include the brute-force, importance-sampling, optimization, blocking, interacting-energy, radial-density, analytic-versus-numerical timing, and pair-distance CSV files. The code repository also contains unit tests for formulas, samplers, statistics, and density utilities.

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